

Random Sparse Covariance Generation

Let p be the number of features and let d be the target average number of nonzero off-diagonal entries per row. We sample

$$M = \text{round}(pd/2)$$

distinct unordered feature pairs uniformly at random. For each selected pair, we independently sample a positive weight from $\text{Uniform}(0.5, 1.5)$. Placing these weights symmetrically produces a sparse symmetric matrix A with zero diagonal. Thus, the average number of nonzero off-diagonal entries per row is approximately d , without imposing a block structure.

For a fixed $0 < \rho < 1$, define

$$\Sigma = I_p + \rho \frac{A}{\|A\|_{\text{op}}}.$$

Because every eigenvalue of $A/\|A\|_{\text{op}}$ lies in $[-1, 1]$,

$$\lambda_{\min}(\Sigma) \geq 1 - \rho > 0.$$

Therefore, every generated matrix is a valid covariance matrix. We accept a draw only when

$$\lambda_1(\Sigma) - \lambda_2(\Sigma) \geq \Delta,$$

where $\Delta > 0$ is specified before the search. If no draw satisfies this condition within the maximum number of attempts, the procedure stops with an error.

All generation, validation, and search functions will be contained in `code/001_find_sparse_covariance_matrix.R`. The main interface will take p , d , ρ , Δ , the maximum number of attempts, and a random seed. It will return the accepted covariance matrix, its leading eigenpairs, the generation settings, and sparsity and eigenvalue diagnostics. The workflow will save this object as `data/raw/random_sparse_covariance.rds`; no other function in the module will perform file input or output.